

The Cluck is Ticking

The **problem** with **expiration dates**

bǎo zhì qī de wèn tí
保质期的问题

Food waste costs the meat industry an estimated 94 billion USD every year

shí wù làng fèi měinián gěi ròu lèi háng yè zào chéng gū jì yì měi yuán de sǔn shī
食物浪费每年给肉类行业造成估计940亿美元的损失。

Inaccurate expiration date labels often mislead consumers

bù zhǔn què de bǎo zhì qī biāo qiān jīng cháng wù dǎo xiāo fèi zhě
不准确的保质期标签经常误导消费者。

Date labels are typically conservative estimates, which can lead to the unnecessary disposal of safe meat or the accidental consumption of spoiled meat

bǎo zhì qī biāo qiān tōng cháng shì bǎo shǒu de gū suàn zhè kě yǐ dǎo zhì ān quán de ròu bèi bù bì yào de diū qì huò zhě bù xiǎo xīn
保质期标签通常是保守的估算，这可以导致安全的肉被不必要地丢弃，或者不小心吃到变质的肉。

Health Consequences of Unsafe Meat

bù ān quán ròu lèi de jiàn kāng wēi hài
不安全肉类的健康危害

Consuming spoiled chicken can lead to infections from bacteria such as Salmonella and E. coli

shí yòng biàn zhì de jī ròu kě néng dǎo zhì shā mén shì jūn yǔ dà cháng gǎn jūn děng xì jūn gǎn rǎn
食用变质的鸡肉可能导致沙门氏菌与大肠杆菌等细菌感染。

It is estimated that 1 in 9 people get sick after ingesting contaminated meat, with 1.52 million dying annually

jù gū jì měi rén zhōng jiù yǒu rén zài shí yòng shòu wū rǎn de ròu lèi hòu shēng bìng měi nián yǒu wàn rén yīn cǐ sǐ wáng
据统计，每9人中就有1人在食用受污染的肉类后生病，每年有152万人因此死亡。

Environmental Impacts

duì huánjìng de yǐngxiǎng
对环境的影响

Environmentally, the meat industry drives deforestation for grazing land, which in turn destroys habitats and reduces biodiversity

cónghuánjìng jiǎo dù lái kàn ròu lèi chǎn yè wèi le huò qǔ fàng mù yòng dì ér tuī dòng sēn lín kǎn fá zhè fǎn guò lái yòu pò huài le
从环境角度来看，肉类产业为了获取放牧用地而推动森林砍伐，这反过来又破坏了
qī xī dì jiǎn shǎo le shēng wù duō yàng xìng
栖息地，减少了生物多样性。

Decomposing chicken releases methane (a greenhouse gas)

fǔ làn de jī ròu huì shì fàng jiǎ wán yī zhǒng wēn shì qì tǐ
腐烂的鸡肉会释放甲烷（一种温室气体）。

Farms release harmful chemicals and waste products that pollute the atmosphere, waterways, and soil

nóngchǎng huì shì fàng yǒu hài huà xué wù zhì hé fèi qì wù wū rǎn dà qì shuǐ dào hé tǔ rǎng
农场会释放有害化学物质和废弃物，污染大气、水道和土壤。

Our project

wǒ men de xiàng mù
我们的项目

Our project is to engineer a biosensor to detect spoilage in meat as an alternative to traditional expiration dates

wǒ men de xiàng mù shì shè jì yī zhǒng shēng wù chuángǎn qì yòng yú jiǎn cè ròu lèi biàn zhì zuò wéi chuántǒngbǎo zhì qī biāoqiān tǐ
我们的项目是设计一种生物传感器，用于检测肉类变质，作为传统保质期标签替
dài fāng àn
代方案。

The biosensor measures cadaverine, an organic compound produced as meat decomposes

gāi shēng wù chuángǎn qì jiǎn cè shǐ àn zhè shì yī zhǒng zài ròu lèi fēn jiě shí chǎnshēng de yǒu jī huà hé wù
该生物传感器检测尸胺，这是一种在肉类分解时产生的有机化合物。

As cadaverine levels rise, the biosensor responds by producing a blue chromoprotein, delivering a clear visual result

suí zhe shǐ àn shuǐ píng de shēng gāo gāi shēng wù chuángǎn qì huì tōngguò chǎnshēng lán sè sè dàn bái lái xiǎng yīng chéng xiàn chū qīng xī
随着尸胺水平的升高，该生物传感器会通过产生蓝色色蛋白来响应，呈现出清晰
de shì jué jié guǒ
的视觉结果。

In the future, we aim to adapt the biosensor for other meats, not just chicken

wèi lái wǒ men de mù biāo shì jiāng gāi shēng wù chuángǎn qì tuī guǎng yìng yòng yú qí tā ròu lèi ér bù jǐn jǐn shì jī ròu
未来，我们的目标是将该生物传感器推广应用于其他肉类，而不仅仅是鸡肉。

Our team

wǒ men de tuán duì
我们的团队

We are the only high school iGEM team in Northern California

wǒ men shì běi jiā zhōu wéi yī de gāo zhōng tuán duì
我们是北加州唯一的高中iGEM团队。

Based at Khan Lab School in Mountain View, including students from nine high schools across the San Francisco Bay Area

wǒ men de jī dì zài shān jǐng chéng de chéng yuán lái zì jiù jīn shān wān qū de jiǔ suǒ gāo zhōng
我们的基地在山景城 (Mountain View) 的 Khan Lab School，成员来自旧金山湾区的九所高中。

About the name

guān yú xiàng mù míng chēng
关于项目名称

"The Cluck is Ticking" is a pun on "the clock is ticking," which means time is running out

"The Cluck is Ticking" 是英文短语 "the clock is ticking" (意为时间紧迫) 的双关语。

The title swaps "clock" for "cluck," the sound a chicken makes, linking the urgency of food safety to our project's focus on chicken

biāo tí bǎ zhōng huàn chéng le jī jiào shēng jiāng shí pǐn ān quán de jīn pò xìng yǔ wǒ men xiàng mù duì jī ròu de guān zhù lián xì le qǐ lái
标题把 "clock" (钟) 换成了 "cluck" (鸡叫声)，将食品安全的紧迫性与我们项目对鸡肉的关注联系了起来。